AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 4 · STUDENT LESSON

Sampling and Variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Fix the draw. Transform the table. Check the interval.

Compare positive-temperature sampling without mistaking consistency for factual correctness.

Keep one habit: Interval contract → exact transformation → two-token picture → paired traces → independent table.

MAIN LESSON · SECTION 1

How does the same draw change under a different distribution?

> Trace sampling boundaries and positive temperature without claiming factual accuracy.

Your learning goals

I can sample from a distribution using explicit **cumulative interval**s.

I can calculate **temperature**-adjusted probabilities for a toy example.

I can explain why random variation and factual reliability need separate evaluation.

Recall greedy versus whole paths

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

- Does the largest next-token probability guarantee the largest complete-path probability?

Your end goal

By the end, I can calculate a toy **temperature** change and trace a **sampling** draw.

▼ Attention extension · Permitted normalised weights

What must happen to permitted weights after excluding a position?

> **My readiness explanation**

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 4: **Sampling** and **Variability**

Decoding choices trade consistency, diversity, and risk.



CREATOR MISSION

Compare low- and high-**variability** story continuations.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Readiness for A small attention calculation

This week’s end goal

MAIN LESSON · SECTION 2

Define sampling intervals before selecting a token

> The left endpoint belongs to the interval; the right endpoint does not.

The complete red/blue/green distribution and interval contract

Start with probabilities red 0.6, blue 0.3, green 0.1 in that fixed order. Greedy chooses red. **Sampling** uses a draw u with 0 <= u < 1 and **cumulative interval**s: red [0,0.6), blue [0.6,0.9), green [0.9,1).

Left endpoints are included, right endpoints excluded. Thus u=0.6 selects blue and u=0.9 selects green. The draw 1 is outside this contract.

Separate repeated frequencies from a single trace

Sampling does not mean choosing all tokens equally. Across many independent draws the frequencies may approach the specified probabilities, but a small sample need not match them exactly.

Fixed recorded draws allow a reproducible classroom trace. A random seed and generator version help replay a real **sampling** program, not establish the truth of **output**s.

▼ **Attention extension · Scores, scale and stable softmax**

For one query q and keys k_i of width d, compute **score** s_i = dot(q,k_i)/sqrt(d). Then obtain weights from **softmax** and combine value vectors. Dot products may be negative; probabilities cannot.

The **softmax** performs the conversion. This is a single-head classroom calculation, not a full language model.

Define **softmax** weights as exp(s_i-m)/sum_j exp(s_j-m), where m is the maximum permitted **score**.

Subtracting the same maximum from every permitted **score** leaves the ratios unchanged and avoids unnecessary overflow from large positive **score**s.

Natural exponent exp is introduced here; use a calculator or Python math.exp rather than estimating its meaning from the word exponential.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **sampling** and **variability** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: A small attention calculation

MAIN LESSON · SECTION 3

Calculate the positive-temperature transformation

> For this half-temperature example, square then normalise.

The stated formula and exact adjusted probabilities

For strictly positive probabilities p_i and **temperature** T>0, define q_i = p_i^(1/T) / sum_j p_j^(1/T). This classroom formula is equivalent to scaling logits before a **softmax** when p came from those logits.

We use it directly on the supplied table. At T=1 the distribution is unchanged. At T=0.5, square each probability: 0.36,0.09,0.01; total 0.46. The adjusted distribution is red 18/23, blue 9/46, green 1/46.

Compare the original and adjusted tables

Token	T=1 probability	T=0.5 probability
red	0.6	18/23 ≈ 0.782609
blue	0.3	9/46 ≈ 0.195652
green	0.1	1/46 ≈ 0.021739

Trace the same draw under both distributions

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T = 1	Intervals 0 to .6 to .9 to 1	Draw .7 selects blue
T = 0.5	First boundary 18/23	Draw .7 selects red
Same recorded draw	Changed distribution	Different output possible

Temperature changes probability concentration; the same draw can fall in a different interval.

Explain the deciding evidence

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Attention extension · Query/key/value roles and exact ratio

Item	Key, width 1	Value, width 1
A	[0]	[2]
B	[ln(3)]	[6]

Take q=[1] and d=1. The scale sqrt(1) is 1, so **score**s are 0 and ln(3). ln(3) is the number whose natural exponential is 3, approximately 1.098612. Exponentials have ratio 1:3, hence weights 1/4 and 3/4.

Output is (1/4)*2 + (3/4)*6 = 5. Using maximum subtraction gives exponentials 1/3 and 1 and the same normalised weights.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
q=[1], keys 0 and ln(3)	Scaled scores 0, ln(3)	Softmax weights .25, .75
Values 2 and 6	Weighted sum .25*2 + .75*6	Output 5
Mask excludes B	Only A permitted	Weight 1 and output 2

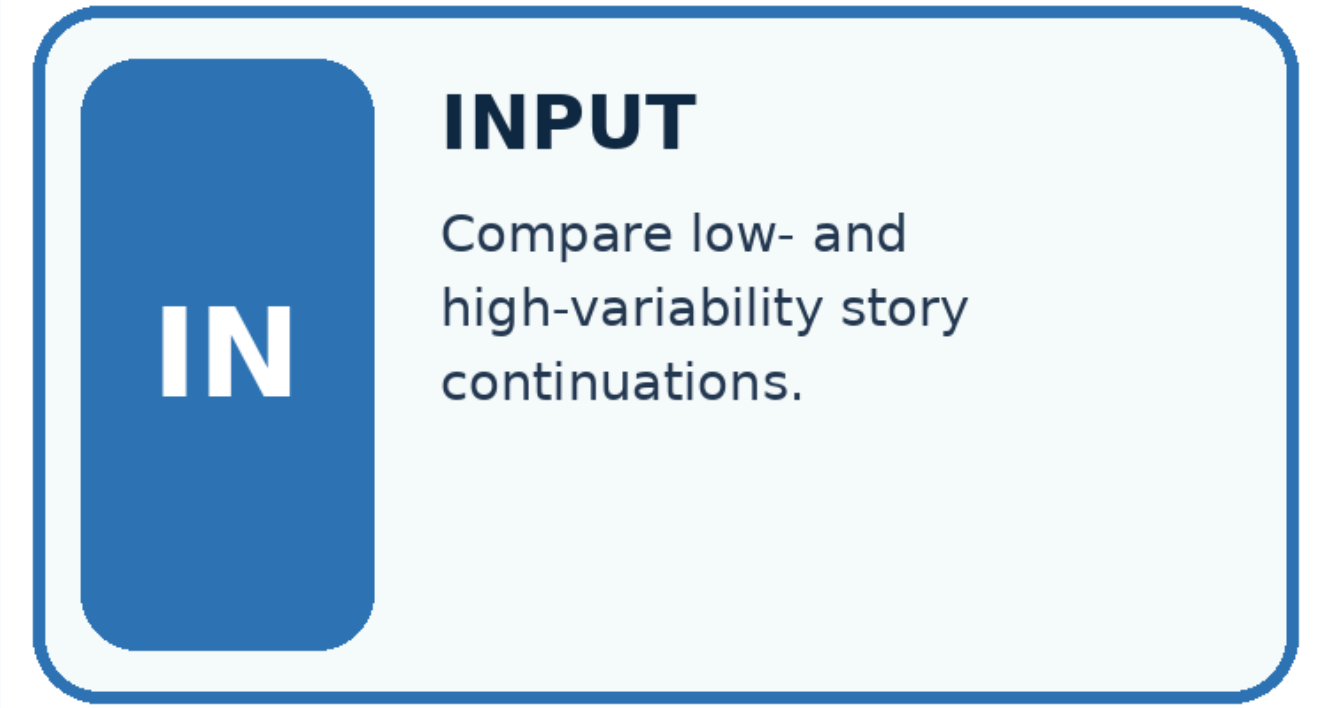
Scores, normalised weights and value combinations are separate stages.

▼ Earlier wording, alternatives and picture transcripts

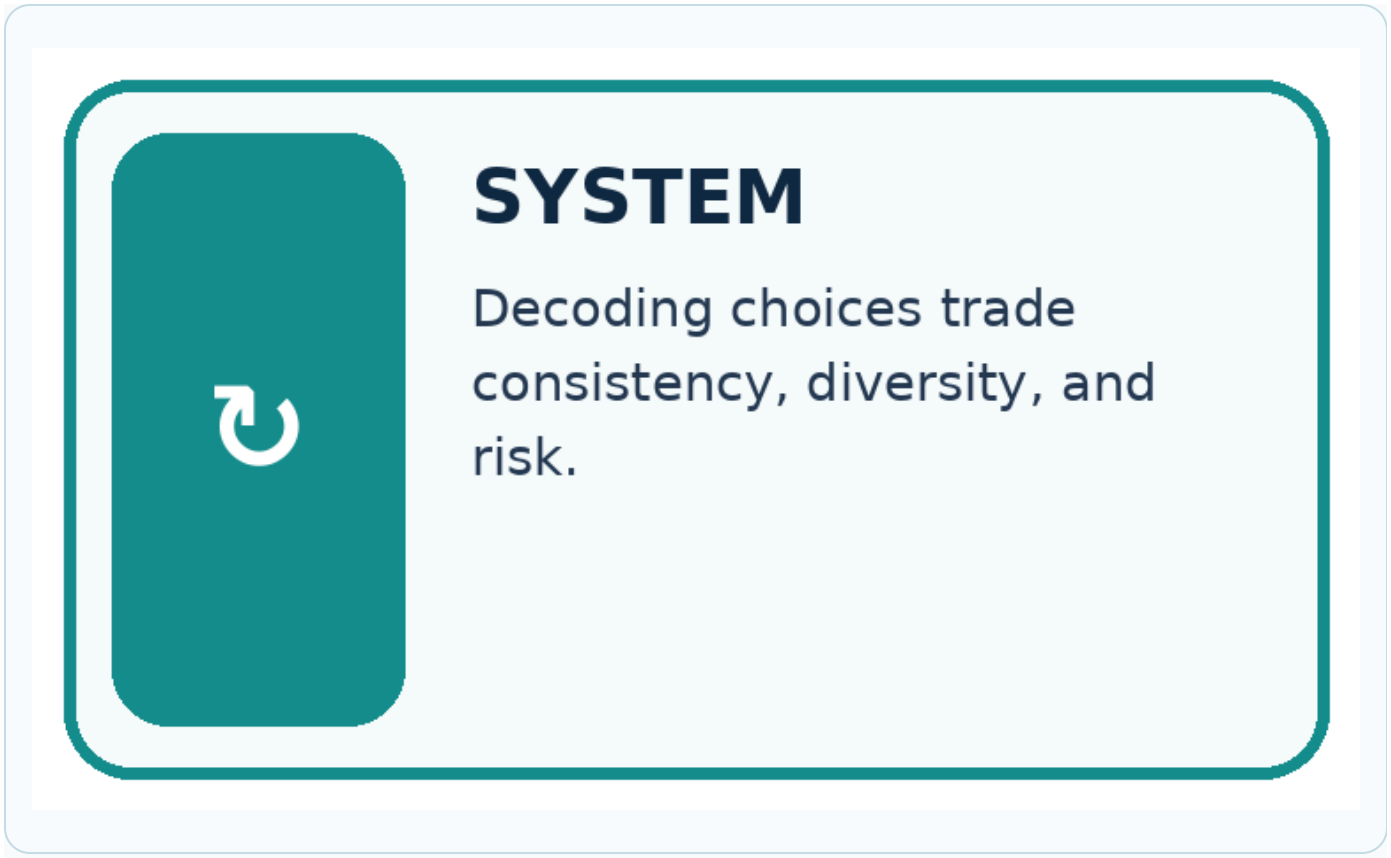
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Decoding choices trade consistency, diversity, and risk.
↓
OUT
EVIDENCE
Compare low- and high-variability story continuations.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: A small attention calculation

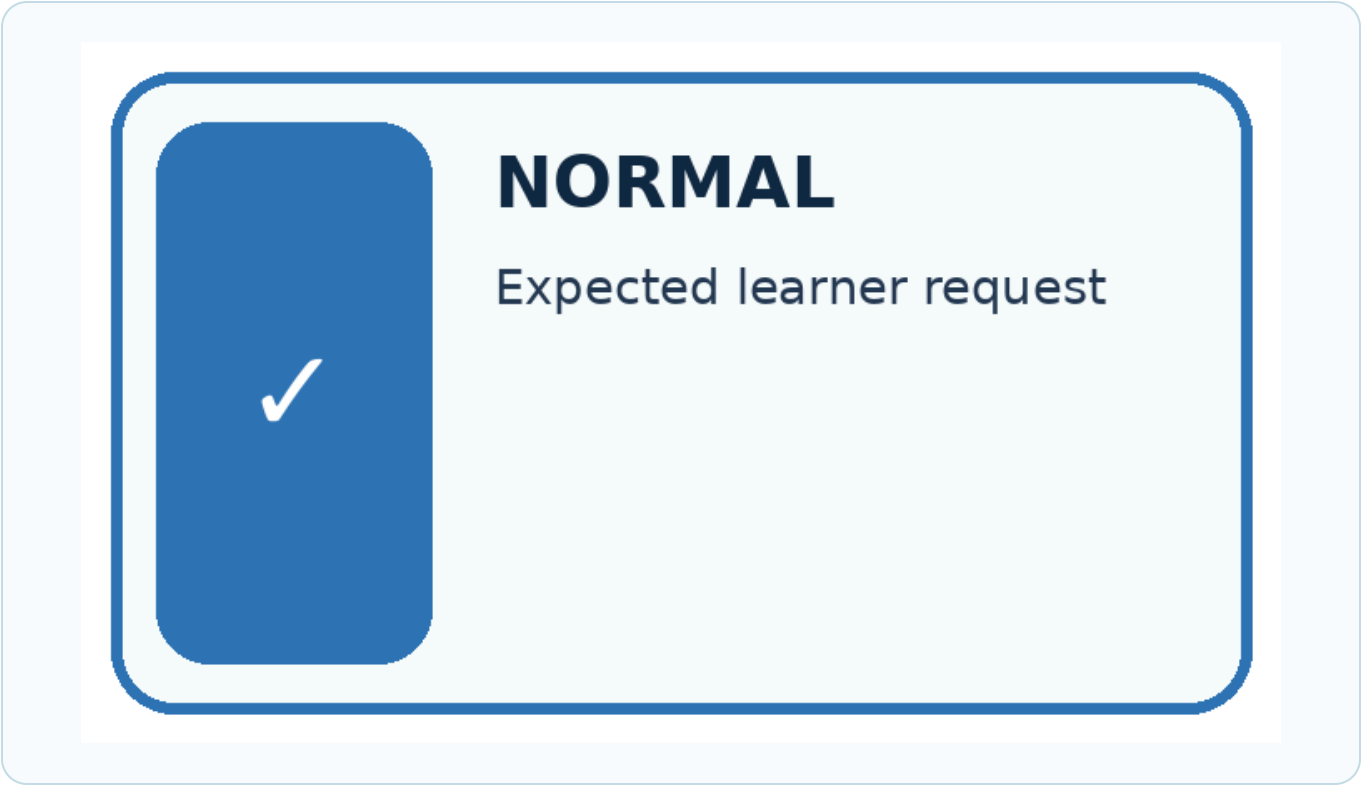
Connect the picture and the explanation

MAIN LESSON · SECTION 4

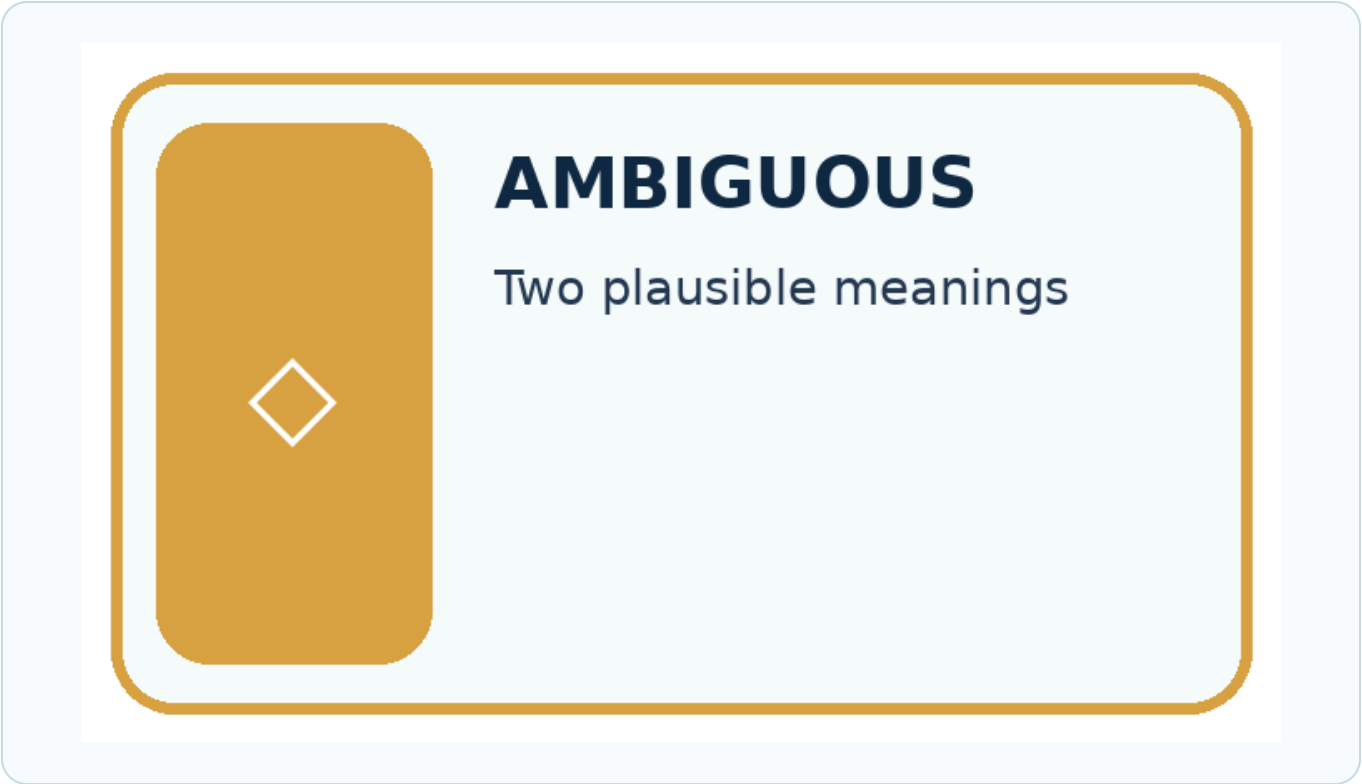
Compare changed intervals with unchanged factual evidence

> A more concentrated distribution is not a source of new truth.

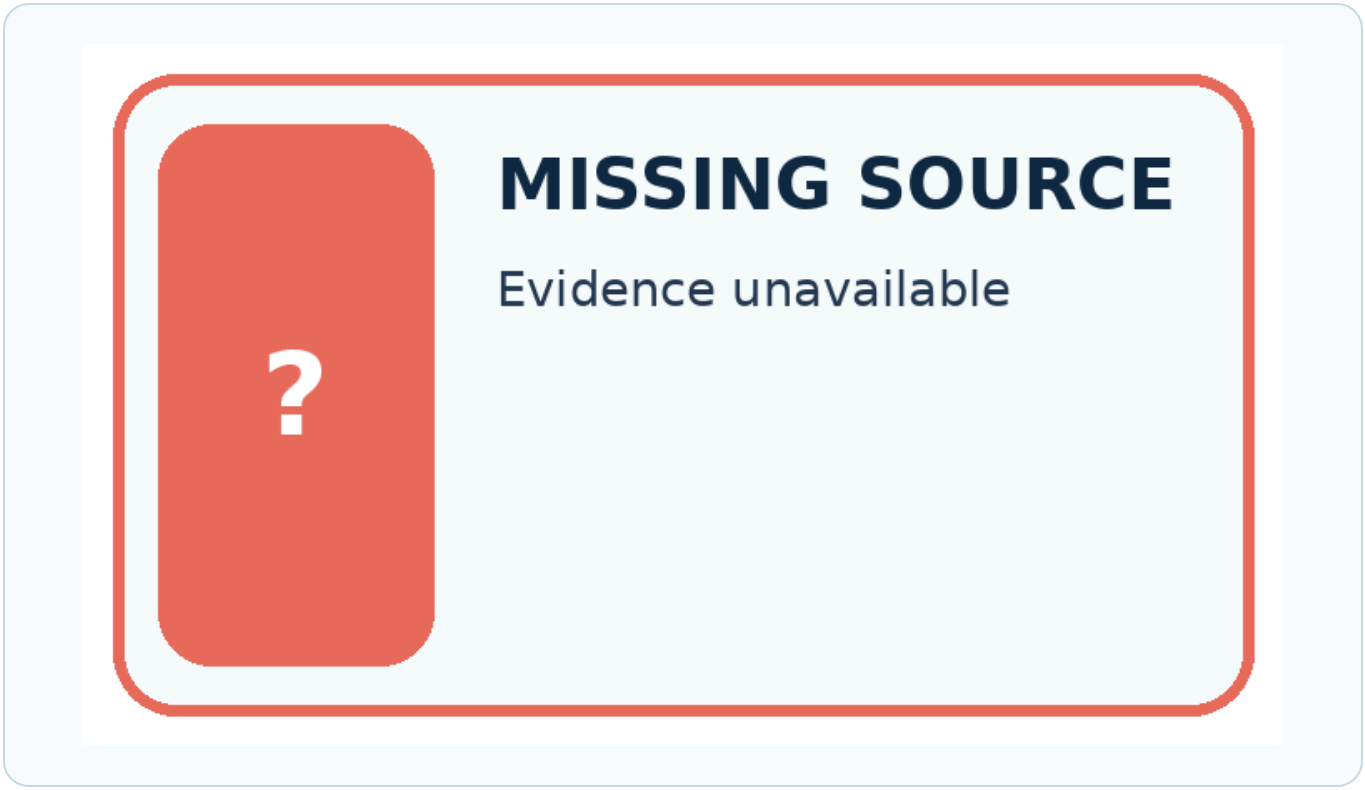
Four pictures with a purpose



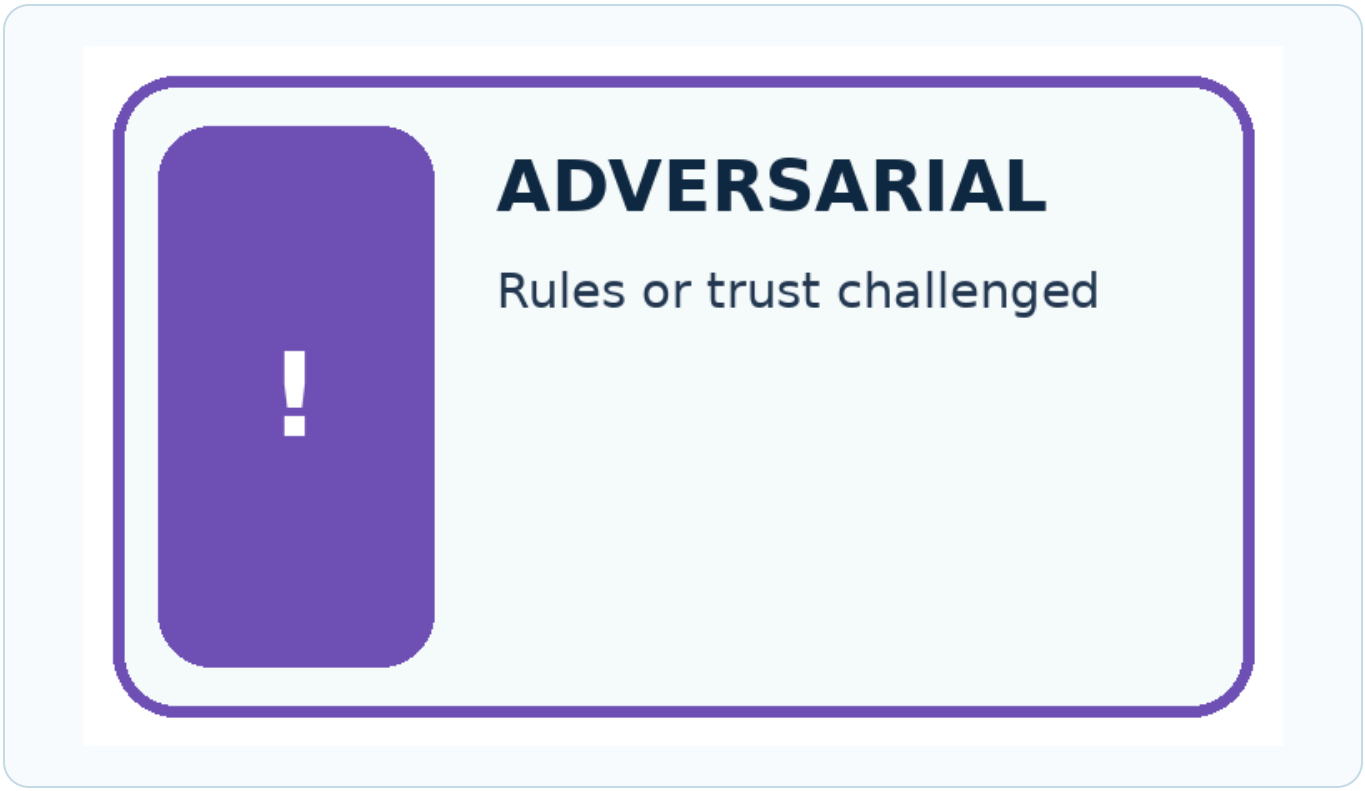
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Set the draw to 0.97, keeping both distributions.

▼ Check after predicting

Prediction: Both **temperature**s select B.

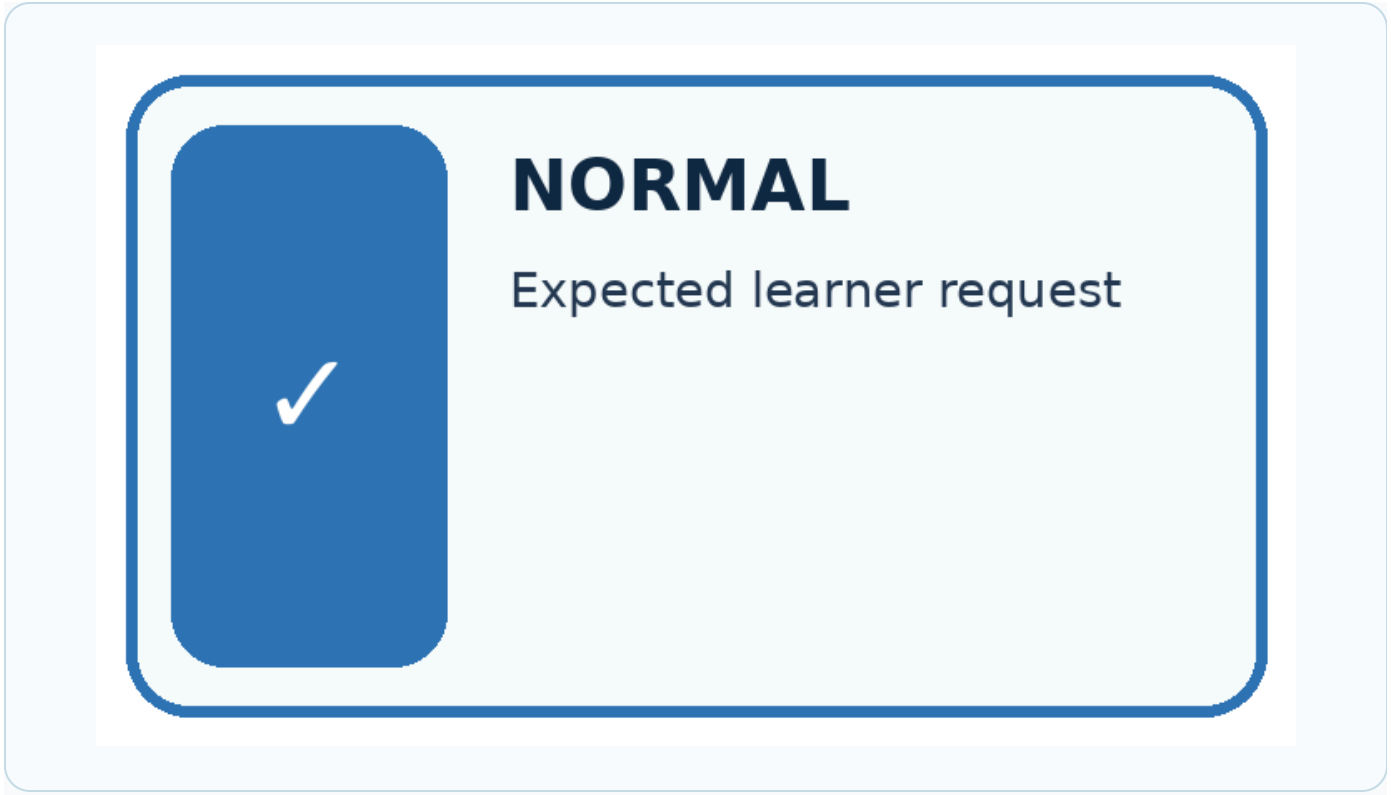
Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

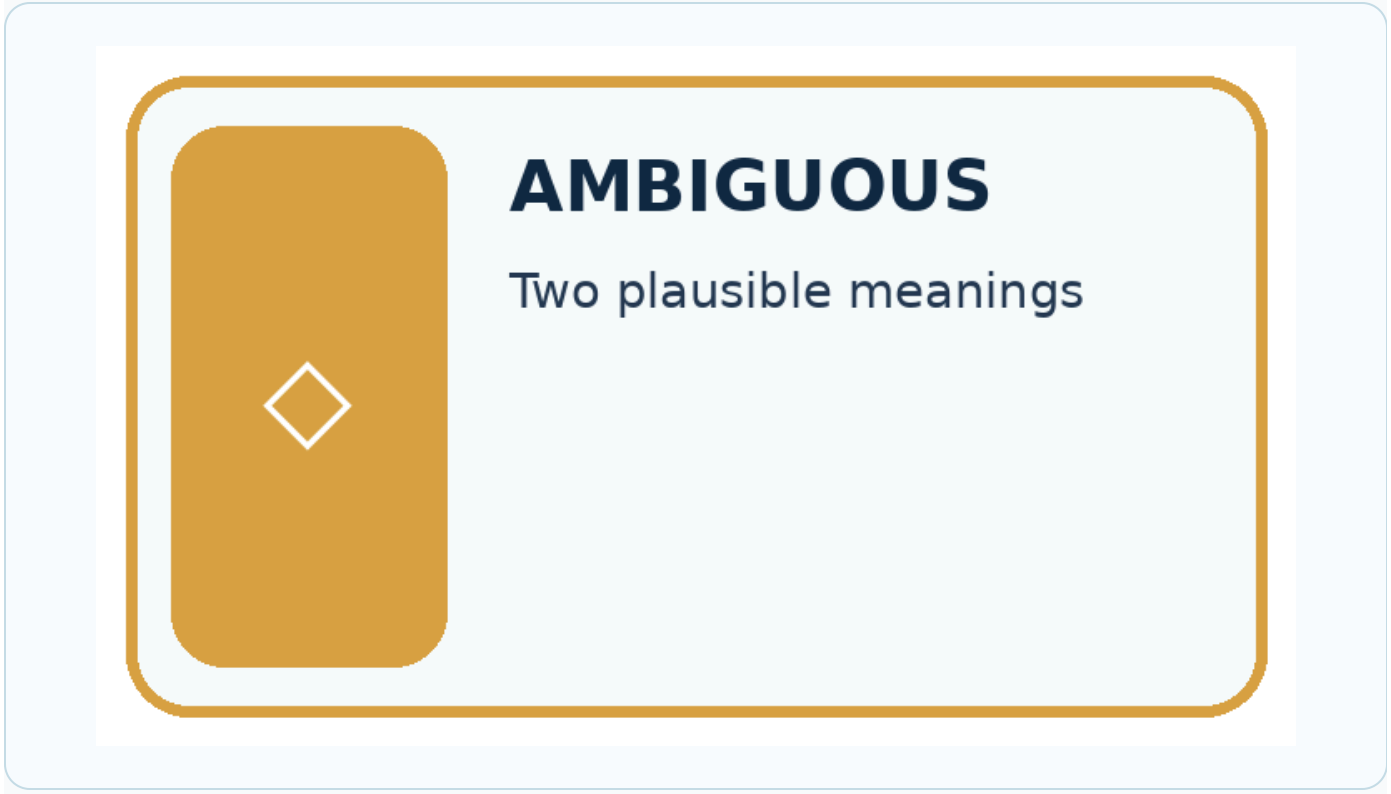
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Explain the temperature limit, then a separate two-token picture

> Keep the three-colour and A/B distributions distinct.

Compare positive temperatures without changing rank

The lower positive **temperature** concentrates this table on its largest entry. For T=2, use square roots before normalising; this flattens this unequal distribution.

These changes leave the order of positive probabilities unchanged, so greedy still selects red. They change the **sampling** distribution, not the factual **evidence** behind a generated claim.

Reject substituting zero into this formula

Our formula excludes T=0; dividing by zero is undefined. Some systems offer a zero-**temperature** setting as a separate greedy policy, but that is not a value to substitute into this formula.

We also avoid claiming that any one setting removes all nondeterminism in an unspecified system.

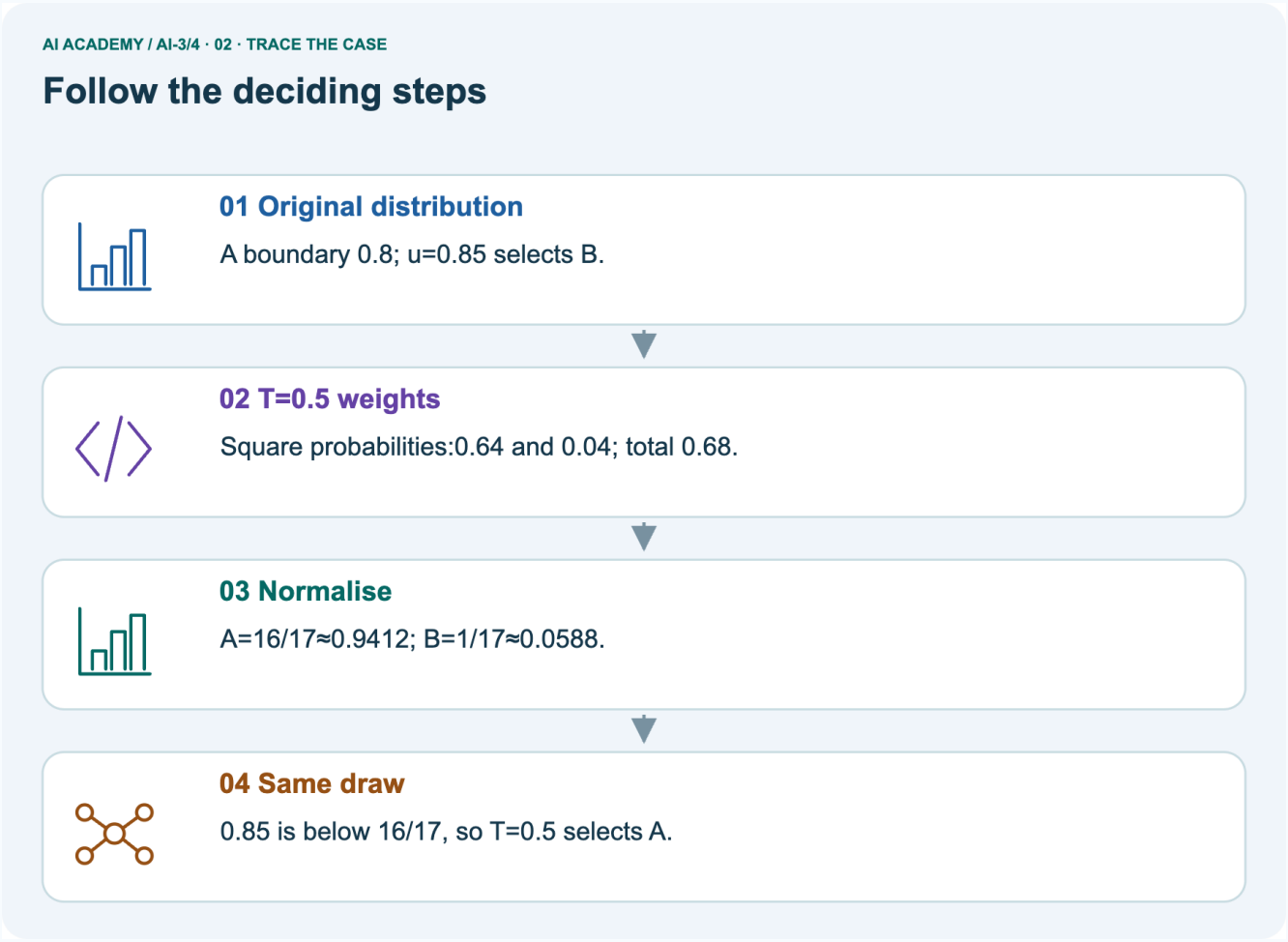
Read the separate two-token distribution and draw

A fictional two-token distribution illustrates **variability** in generation.

At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

Trace squared weights and exact intervals



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Original distribution	A boundary 0.8; u=0.85 selects B.
2. T=0.5 weights	Square probabilities:0.64 and 0.04; total 0.68.
3. Normalise	A=16/17≈0.9412; B=1/17≈0.0588.
4. Same draw	0.85 is below 16/17, so T=0.5 selects A.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

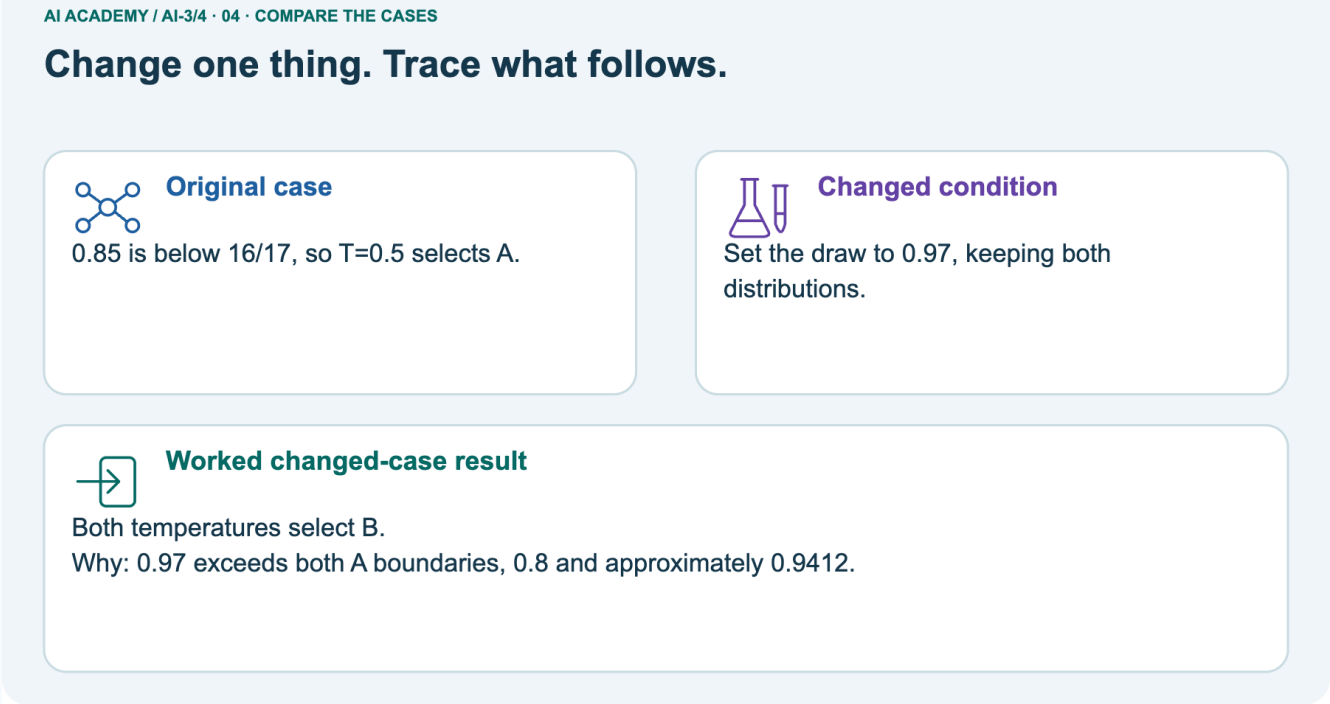
Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Predict the changed draw0.97

- Change one thing: Set the draw to 0.97, keeping both distributions. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain remaining probability and truth limits

- Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.
- Read the labelled picture: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

▼ Attention extension · Equal scores and masks

For q=[1,1], kA=[1,0] and kB=[0,1], both dot products are 1. Divide by sqrt(2); equal **score**s give equal weights 1/2 regardless of their common value. With values [2,0] and [0,4], the **output** is [1,2].

Do not omit the scale merely because it cancels in this tied example; unequal **score**s generally change concentration when scaled.

Apply masks before normalising over permitted **score**s. In the width-one example, excluding B leaves weight 1 for A and **output** 2. A key with the wrong width violates the dot-product contract.

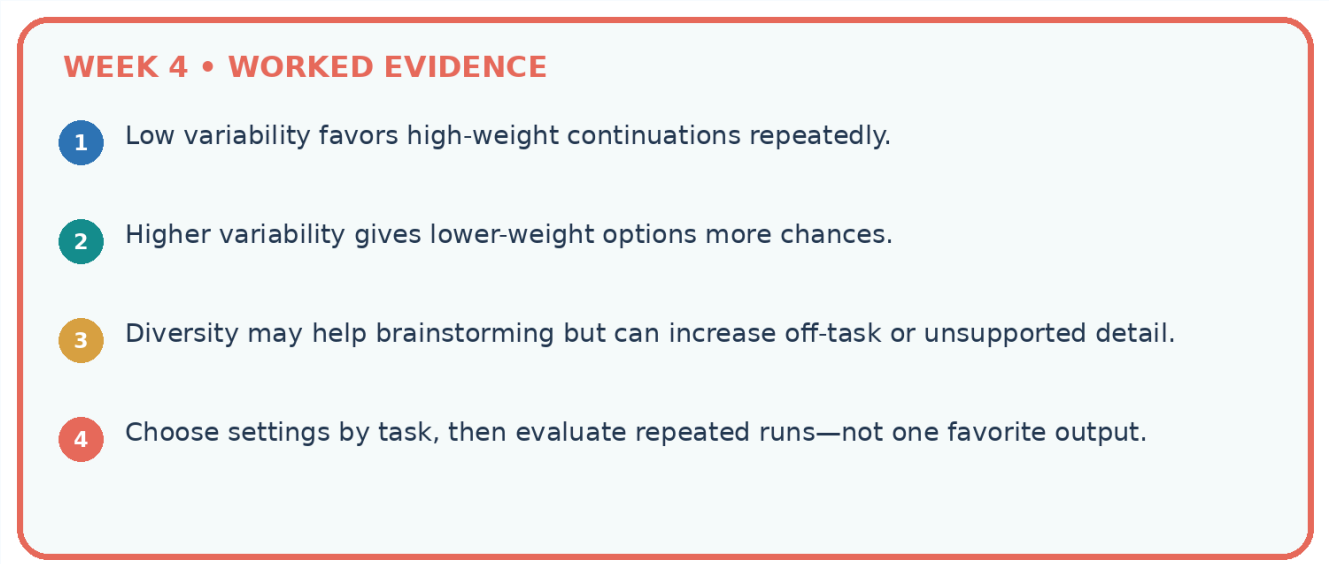
Values must share a width with one another, but their width need not equal the query/key width. An all-excluded mask is handled explicitly instead of taking a maximum of an empty list.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Low **variability** favors high-weight continuations repeatedly.

2. Higher **variability** gives lower-weight options more chances.

3. Diversity may help brainstorming but can increase off-task or unsupported detail.

4. Choose settings by task, then evaluate repeated runs—not one favorite **output**.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Apply the connected idea: A small attention calculation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling and Variability

The facts we will use

AI ACADEMY / AI-3/4

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.058823529411764705

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/4

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Original distribution

A boundary 0.8; u=0.85 selects B.



2. T=0.5 weights

Square probabilities:0.64 and 0.04; total 0.68.



3. Normalise

A=16/17≈0.9412; B=1/17≈0.0588.



4. Same draw

0.85 is below 16/17, so T=0.5 selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Separate consistency from correctness

> Inspect factual support with its own reference rather than inferring it from concentration.

Repair the lower-temperature truth claim

Claim to repair: Lower temperature made red more likely, so red is more accurate. The operation transforms probabilities and sampling intervals. It does not supply evidence about the actual colour. A highly consistent wrong token remains wrong. In the two-token picture B retains probability1/17 at T=0.5; less likely is not impossible, and u=0.97 still selects B.

Design a revealing fixed-draw comparison

> **Exit: a team lowers temperature and claims hallucinations are impossible. State what was changed and what evidence that claim still lacks.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Higher variability makes a model more knowledgeable. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

A fair **temperature** comparison fixes the prefix, model distribution, candidate order and draw policy. Record multiple outcomes and evaluate task criteria separately.

If the aim is creativity in fiction, variation may be useful; if the aim is reporting a sensor state, every factual claim still needs **evidence**.

Never call a probability-concentration change an accuracy improvement without an appropriate evaluation.

Apply the connected idea: A small attention calculation

Check query/key dimensions, scale, permitted positions, finite **score**s, stable normalisation, weight sum and value widths in that order. A zero **output** can result from values cancelling; it does not imply zero attention weights.

Record enough intermediate values for someone else to reproduce the trace.

> **Exit: two equal scores yield weights 1/2,1/2 and values 4,-4 yield output 0. Explain why zero output does not mean neither position was used.**

MAIN LESSON · SECTION 8

Practise exact boundary draws with less help

> Keep exact fractions when a draw lies near an endpoint.

Read and trace the three supplied draws

- Supported draws 0.2,0.6,0.9 at T=1 select red, blue, green. At T=0.5, 0.2 and 0.6 fall below 18/23 and select red; 0.9 lies between 18/23 and 45/46 and selects blue.

  Keep exact fractions for boundaries, because rounding can misclassify a draw near an endpoint.

> **Guided: trace those three draws under both distributions and explain why the result is a procedure trace, not a frequency estimate from a large experiment.**

Complete main temperature W2

Practise with less help: Compute the adjusted table

Use workbook W2 for the connected example "**Sampling** and **temperature**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair sampling means equal chances

- Use the worked case for **Sampling** and **temperature**. Proposed explanation: "**Sampling** means equal chances". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate A/B picture

- Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

Explain the picture

> **My explanation using a pictured step and the given facts**

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Attention extension · Stable common-shift calculation

Supported **score**s 1000 and 1000+ln(3) would overflow an ordinary direct exponential calculation in many floating-point implementations. Maximum subtraction gives -ln(3) and 0, producing weights 1/4 and 3/4.

Their common shift does not change the relative weights. This is a stability **mechanism**, not **evidence** about source correctness.

> **Guided: calculate the shifted scores and weights, then the output for values 4 and 8.**

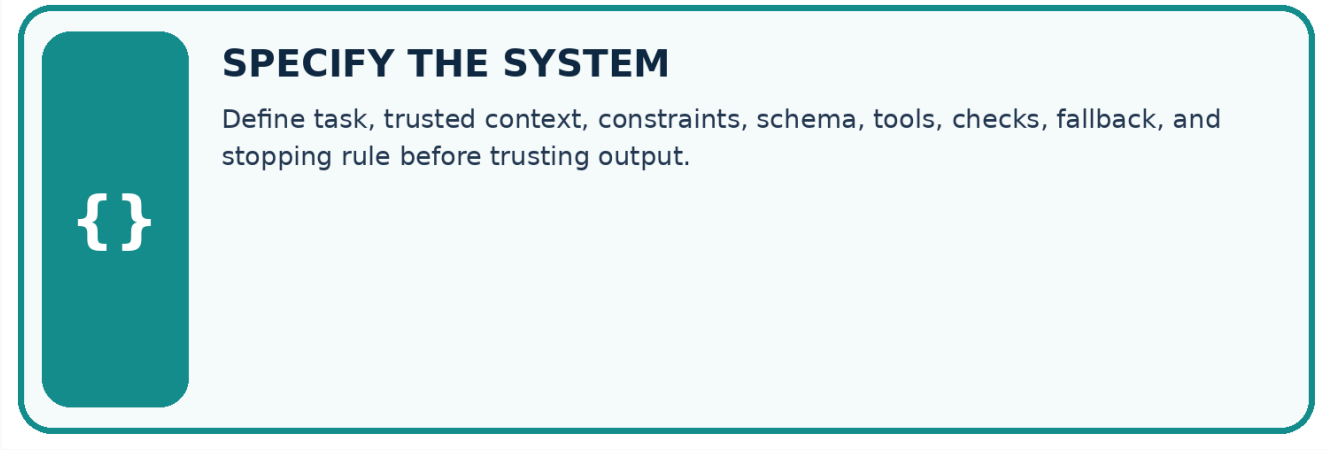
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: **Score**s already are probabilities

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: A small attention calculation

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Practise the connected topic: A small attention calculation

Run a complete fixed-draw comparison

> Hold prompt, candidates and draws fixed while changing the stated distribution transformation.

Predict the controlled setup

1. Complete controlled paper/local experiment: fixed prefix Next colour:, token order red/blue/green, original distribution0.6/0.3/0.1, draws0.1,0.4,0.7,0.85,0.97. Compare T=1 and the stated T=0.5 transformation using the same draws. Predict all outputs and category counts, then run the supplied program or make an explicitly labelled paper trace.

   These are paired deterministic traces of recorded draws, not five new random trials or actual language-model stories.

Run the complete exact-arithmetic simulation

```
from fractions import Fraction as F

def half_temperature(p):
    if not p or any(x<=0 for x in p) or sum(p)!=1:
        raise ValueError('positive probabilities summing to one required')
    weights = [x*x for x in p]
    return [x/sum(weights) for x in weights]

def sample(p,u):
    if not p or any(x<0 for x in p) or sum(p)!=1 or not 0<=u<1:
        raise ValueError('invalid distribution or draw')
    cumulative=F(0)
    for index,probability in enumerate(p):
        cumulative+=probability
        if u<cumulative:
            return index
    raise AssertionError('complete exact distribution must select an interval')

labels=['red','blue','green']
p=[F(6,10),F(3,10),F(1,10)]
q=half_temperature(p)
print('T=0.5:',q)
for u in [F(2,10),F(6,10),F(9,10)]:
    print(u,labels[sample(p,u)],labels[sample(q,u)])
```

```
draws=[F(1,10),F(4,10),F(7,10),F(85,100),F(97,100)]
for name,distribution in [('T=1',p),('T=0.5',q)]:
    outputs=[labels[sample(distribution,u)] for u in draws]
    print(name,outputs,{label:outputs.count(label) for label in labels})
```

Compare consistency with separate supplied evidence

1. For a separate factual check, supplied fictional reference R says the current flag is blue. Score each generated colour label against that reference. Both five-draw traces have two blue outputs, even though the adjusted distribution produces red more often.

   The experiment demonstrates changed sampling behavior and a distinction between consistency and reference agreement; it does not establish an optimal model setting, general factual accuracy or educational benefit. Keep the independent A/B/C table withheld.

Keep the investigation record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied fixed prefix, three-colour table and five recorded draws with the complete local simulation. Compare output counts and agreement with fictional blue-flag reference R separately. These are toy paired traces, not actual prompt runs, stories or model-quality measurements.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: transform the fresh A/B/C table

> Trace both distributions before interpreting the result.

Independent table and scope

Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

Use exact fractions for squared weights and cumulative boundaries. Intervals are left-inclusive,right-exclusive in A,B,C order. Zero temperature is outside the stated formula. The task provides no factual reference for these labels, so explain the procedural comparison without calling either output true.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Sample a new table independently

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use low **variability** for structured study answers and allow more diversity only for labeled brainstorming.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Treat a generation setting as a policy to evaluate

> Keep the source checks regardless of observed variability.

This week’s concrete contribution

For the Verified Study Buddy, record generation settings and keep unrelated settings fixed during a comparison. Evaluate supported facts, response format and task usefulness separately from output similarity. A lower-variability policy may be a hypothesis for a structured task, but the project must test its own criteria; temperature alone does not verify factual claims.

The classroom simulation supplies no recommendation about a deployed service’s best settings.

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Use low **variability** for structured study answers and allow more diversity only for labeled brainstorming.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

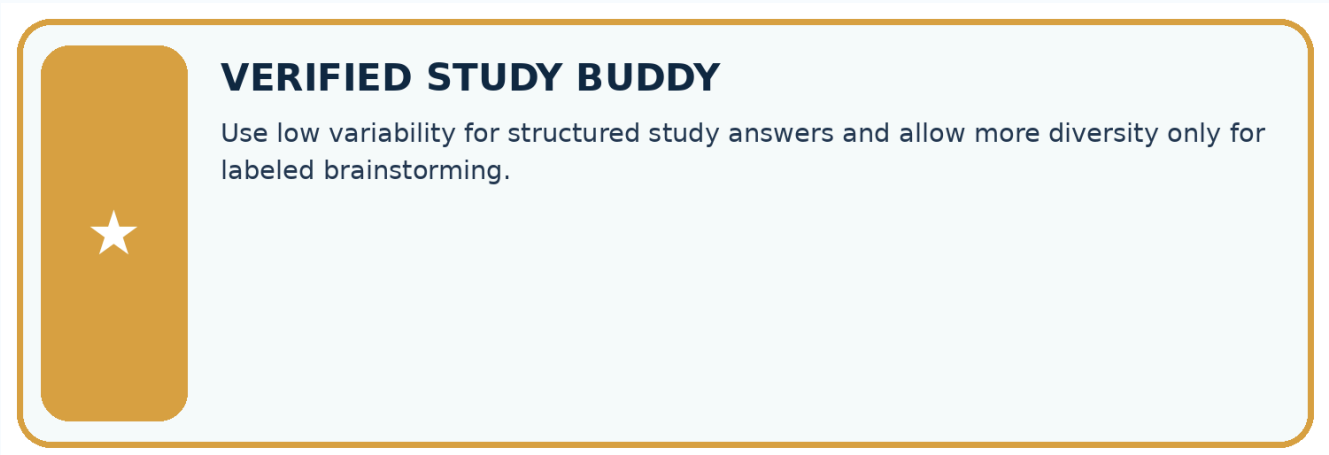
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: explain the interval, transformation and truth boundary

> A fixed-draw trace makes the numerical decision reproducible.

Explain without notes

- Close your notes. Explain this goal in your own words: Sample from a distribution using explicit **cumulative interval**s

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
decoding	Selecting output tokens from a model's scores using a specified procedure.	
variability	Differences observed across repeated runs, samples or conditions.	
temperature intuition	A rough understanding of how temperature rescales token scores: higher values commonly spread choices more, without guaranteeing creativity or truth.	
seeded test	A test containing a deliberately planted, known condition or error to check whether it is handled.	

Main explanation terms

Word	Meaning
Sampling	Selecting an alternative using a specified random procedure

Word	Meaning
Greedy choice	Selecting an alternative with largest next-token probability under a tie rule
Temperature	A positive parameter that changes the concentration of a distribution under a stated transformation
Cumulative interval	A range of draws assigned to an alternative using accumulated probability mass

Extension terms

Term	Meaning
Scaled dot product	A query-key dot product divided by the square root of key width
Softmax	Exponentiating scores and normalising their sum to obtain weights
Score	An unnormalised compatibility value before softmax
Numerical stability	An implementation choice that avoids avoidable arithmetic overflow or loss

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

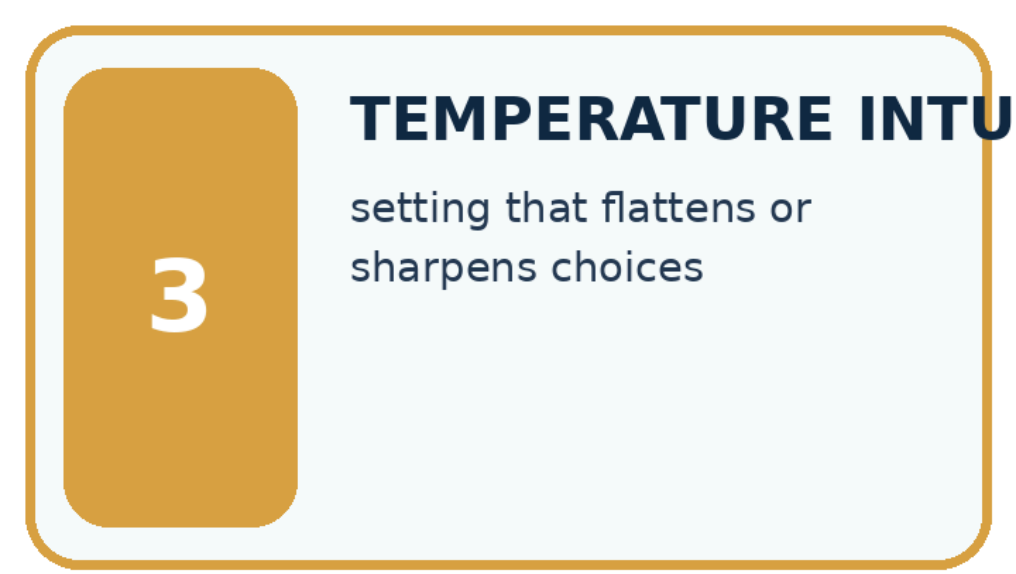
DECODING

method for choosing output tokens

2

VARIABILITY

how much repeated outputs differ





Words used in the closer explanation

Terms for A small attention calculation

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

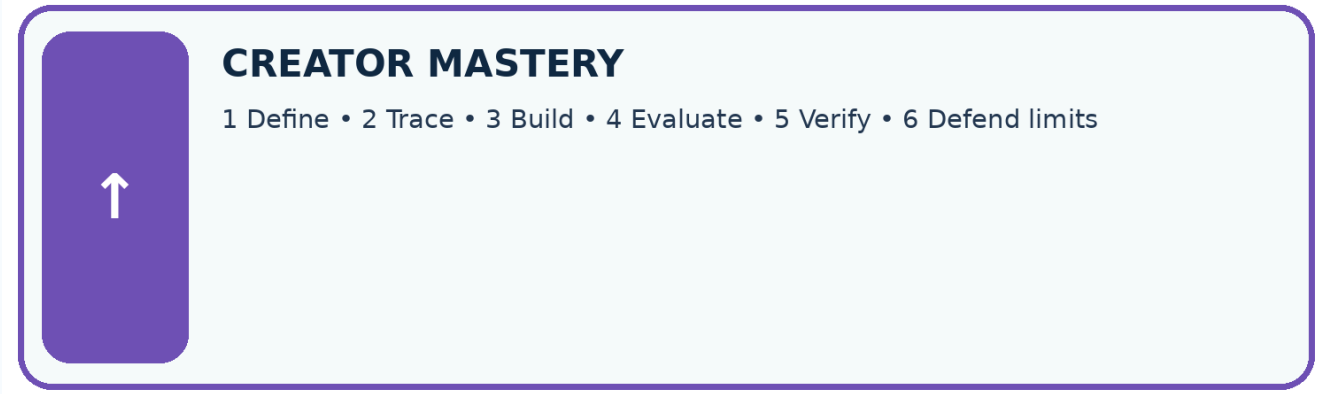
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

REFERENCE · SECTION 12

Reference · Check your explanation

> Use after practice; Sample a new table independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check

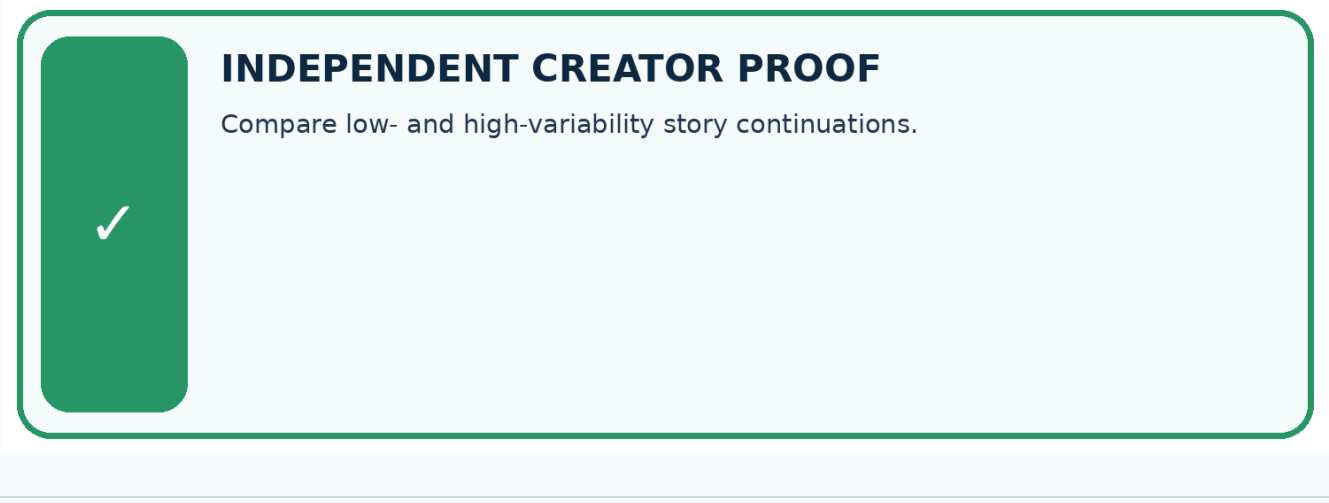
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for this week's explanation

Use after practice. Sample a new table independently is the selected independent task.



Require exact adjusted probabilities, interval endpoints and a separate factual-evidence limit.

▼ **Extra practice · The earlier complete case**

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

One prompt produces three different valid quiz questions.

Does **variability** mean failure?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ **Check after trying**

No. **Sampling** can vary **output**s. **Score** correctness, relevance and format over repeated trials; consistency differs from quality.

Evidence or stage	Value or result
Trial A	Different valid quiz question
Trial B	Another valid quiz question
Trial C	Check correctness, relevance and format

▼ Earlier wording, alternatives and picture transcripts

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Variability and quality are separate

One prompt produces three different valid quiz questions.

Does **variability** mean failure?

No. **Sampling** can vary **output**s. **Score** correctness, relevance and format over repeated trials; consistency differs from quality.

▼ Reference · Earlier checklist

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Sample from a distribution using explicit cumulative intervals	Explain the deciding step, show a trace or test, and state one limit.
Calculate temperature-adjusted probabilities for a toy example	Explain the deciding step, show a trace or test, and state one limit.
Explain why random variation and factual reliability need separate evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional two-token distribution illustrates **variability** in generation.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional two-token distribution illustrates **variability** in generation.
- Which labelled step matches this detail? 0.85 is below 16/17, so T=0.5 selects A.
- 1. Normalise
- 2. Original distribution
- 3. T=0.5 weights
- 4. Same draw
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after trying**

After your attempt, compare with the supported explanation: Same draw 0.85 is below 16/17, so T=0.5 selects A. Lower positive **temperature** concentrates this unequal distribution on its larger entry.

It changes **sampling** probabilities, not **evidence** for whether A or B is true. Using the same draw isolates how the distribution changes the decision.

This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Changed-case question

- Changed-case practice: Set the draw to 0.97, keeping both distributions. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Both **temperature**s select B.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: Both **temperature**s select B.
0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Repair and keep evidence

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

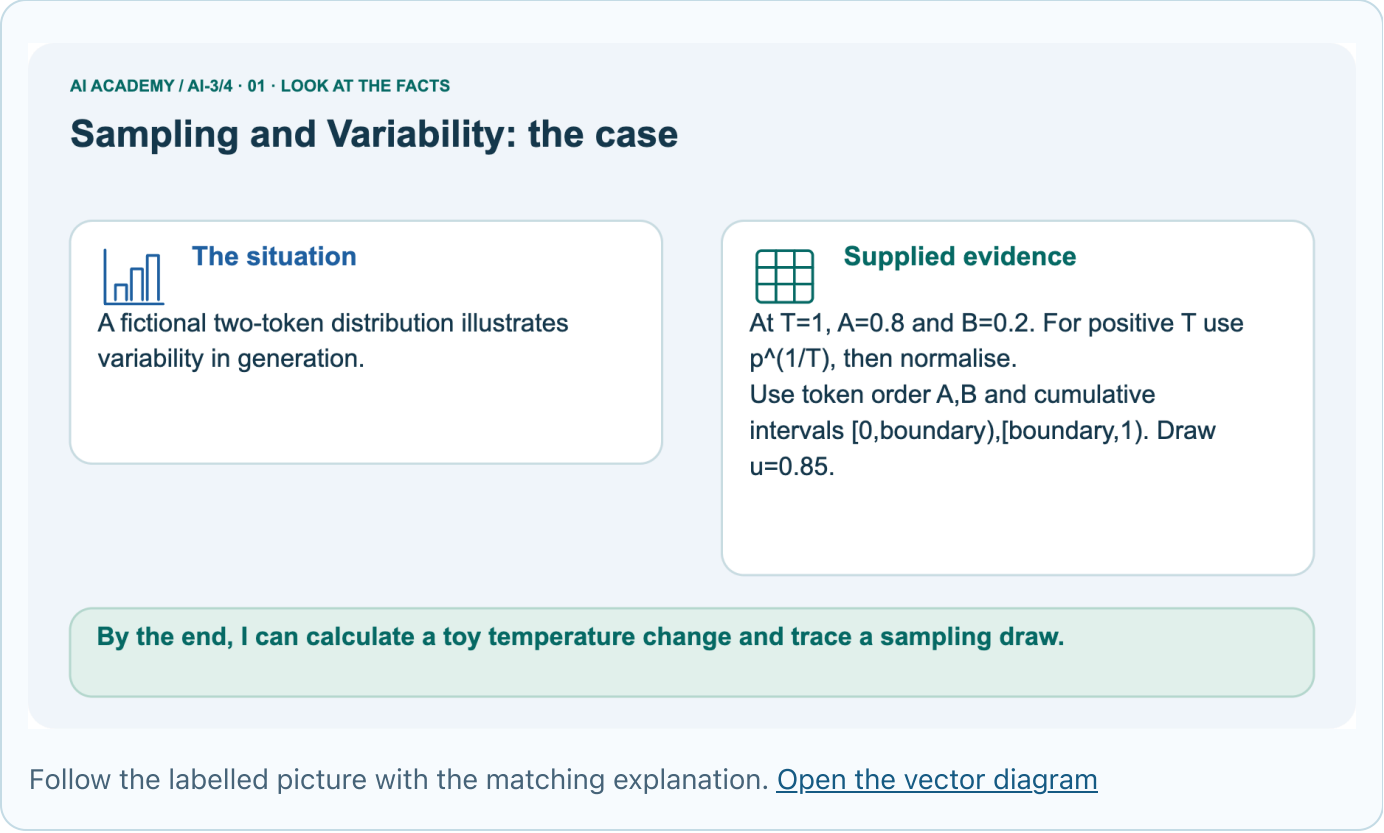
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Sampling and Variability: the case



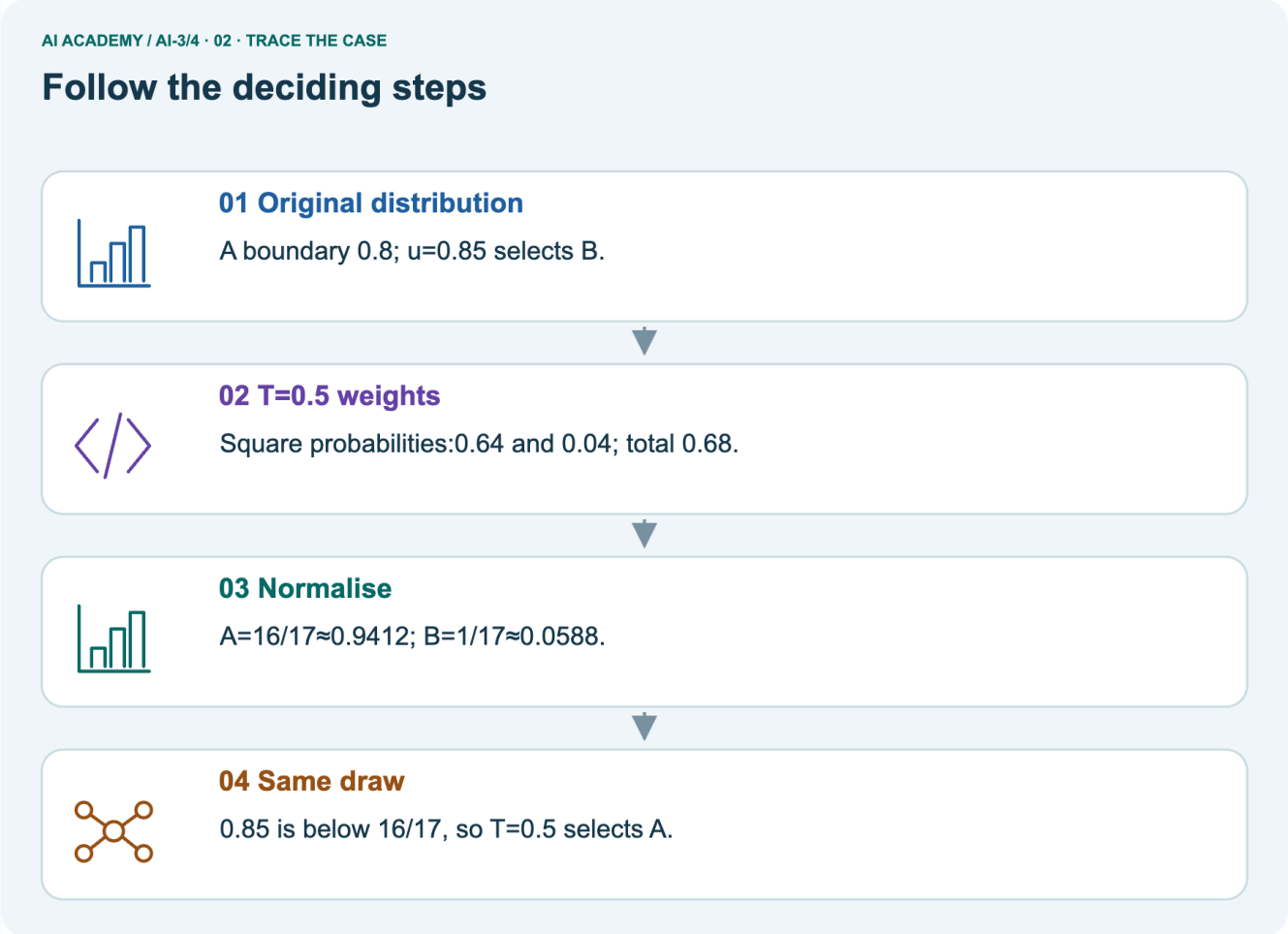
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional two-token distribution illustrates **variability** in generation. Supplied facts: At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

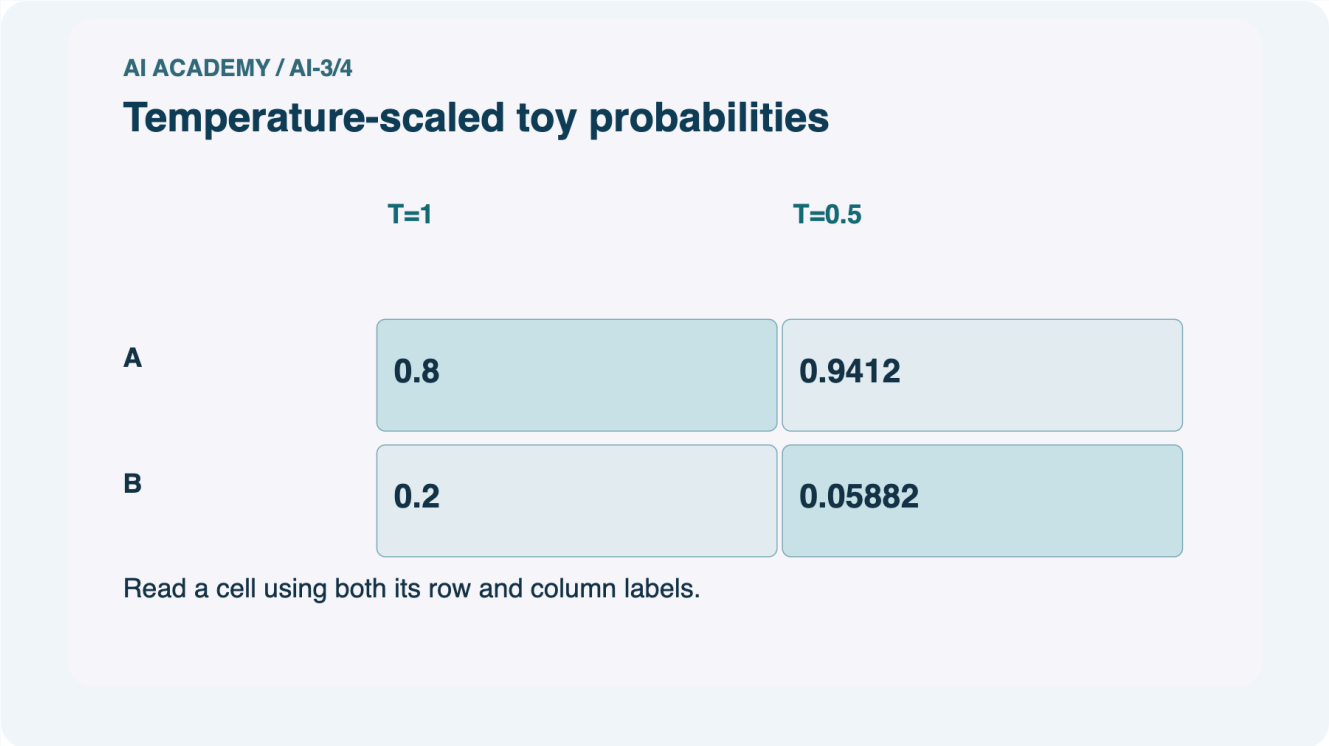


Follow the labelled picture with the matching explanation. Open the vector diagram

Original distribution: A boundary 0.8; u=0.85 selects B. → T=0.5 weights: Square probabilities:0.64 and 0.04; total 0.68. → Normalise: A=16/17≈0.9412; B=1/17≈0.0588. → Same draw: 0.85 is below 16/17, so T=0.5 selects A.

Point to each numbered stage and explain which recorded **input** it uses.

03. Temperature-scaled toy probabilities

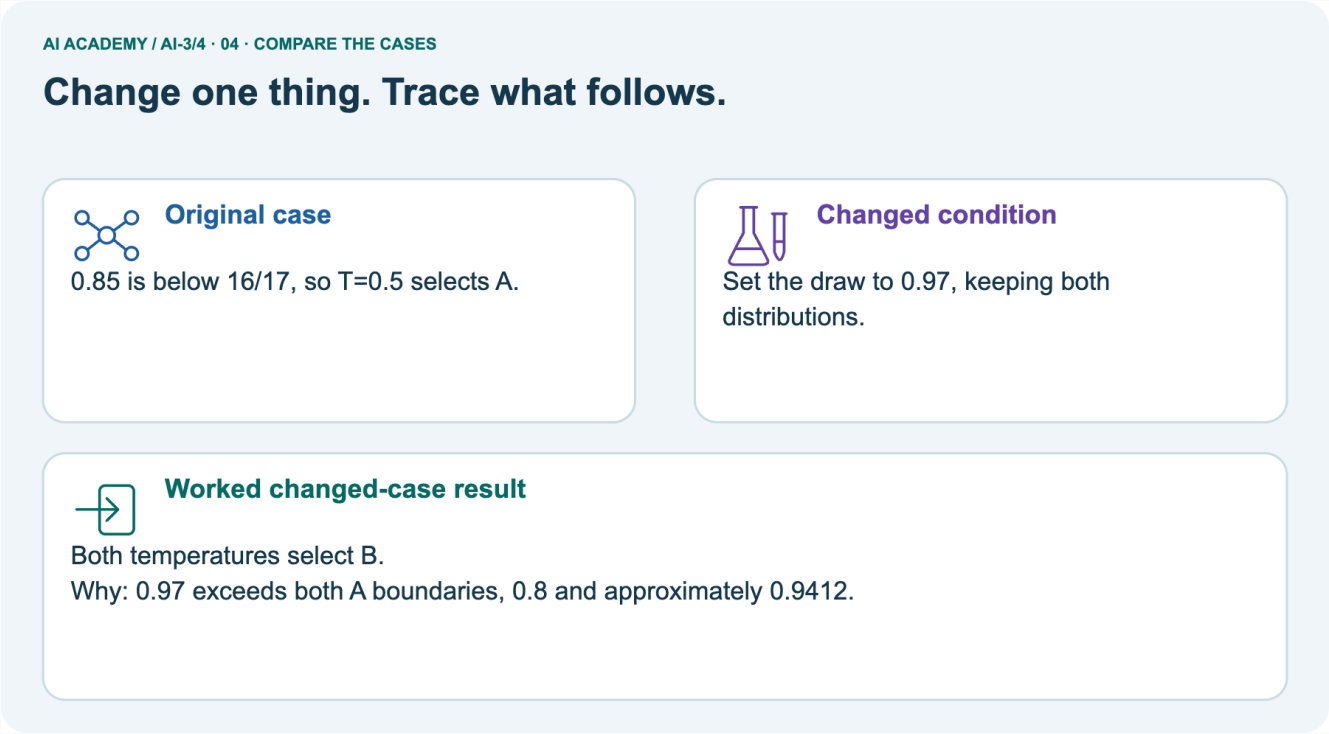


Follow the labelled picture with the matching explanation. Open the vector diagram

Temperature-scaled toy probabilities. A: T=1 = 0.8, T=0.5 = 0.9411764705882353. B: T=1 = 0.2, T=0.5 = 0.058823529411764705.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



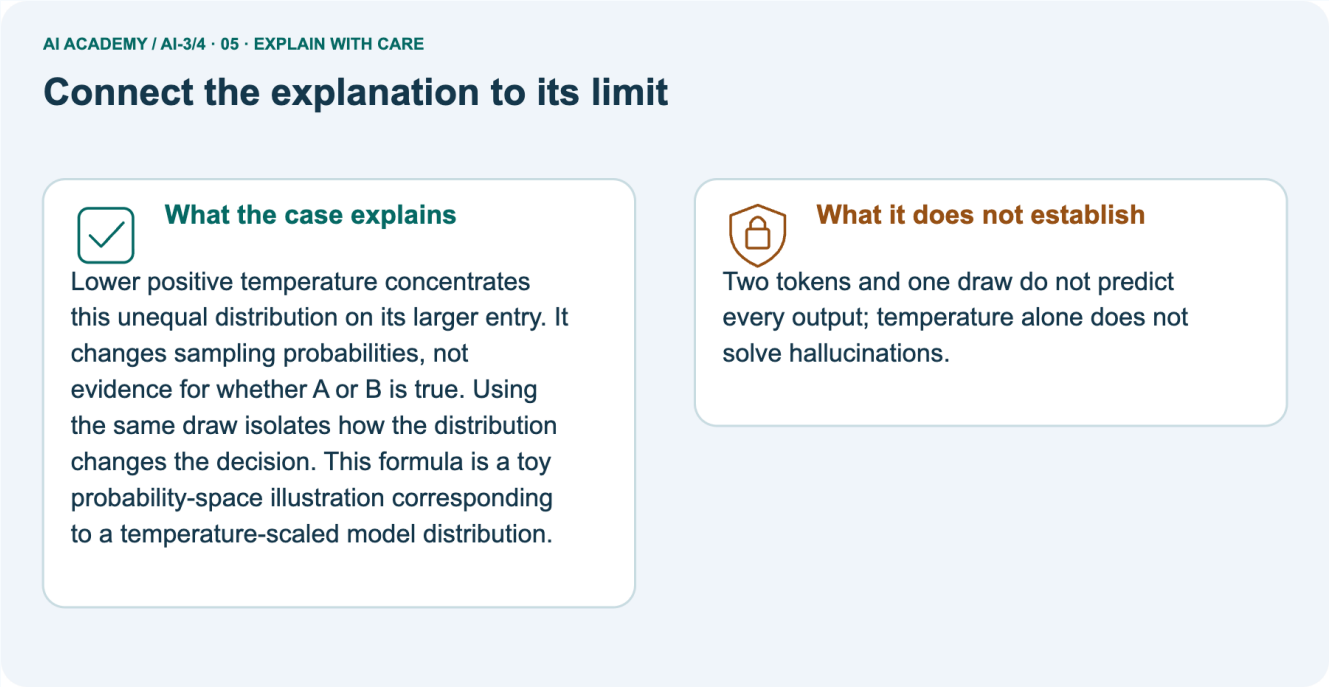
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 0.85 is below 16/17, so T=0.5 selects A. Changed condition: Set the draw to 0.97, keeping both distributions. Expected result: Both **temperature**s select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

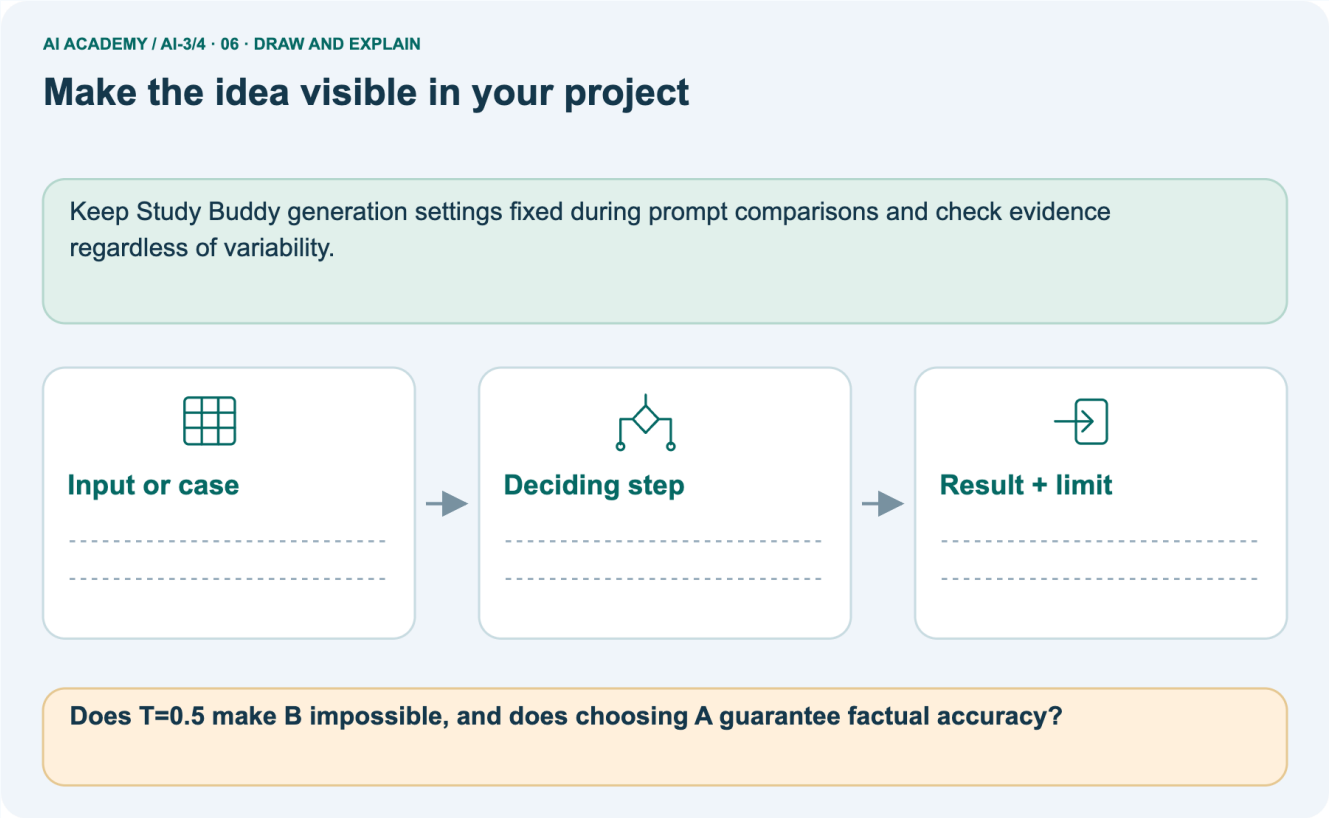
Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Limit: Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**. Draw your **input**, deciding step, result and limit.

Supported question: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Sampling and Variability: the case

02. Follow the deciding steps

03. Temperature-scaled toy probabilities

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.